



Module 3 : Pragmatic & Discourse Processing

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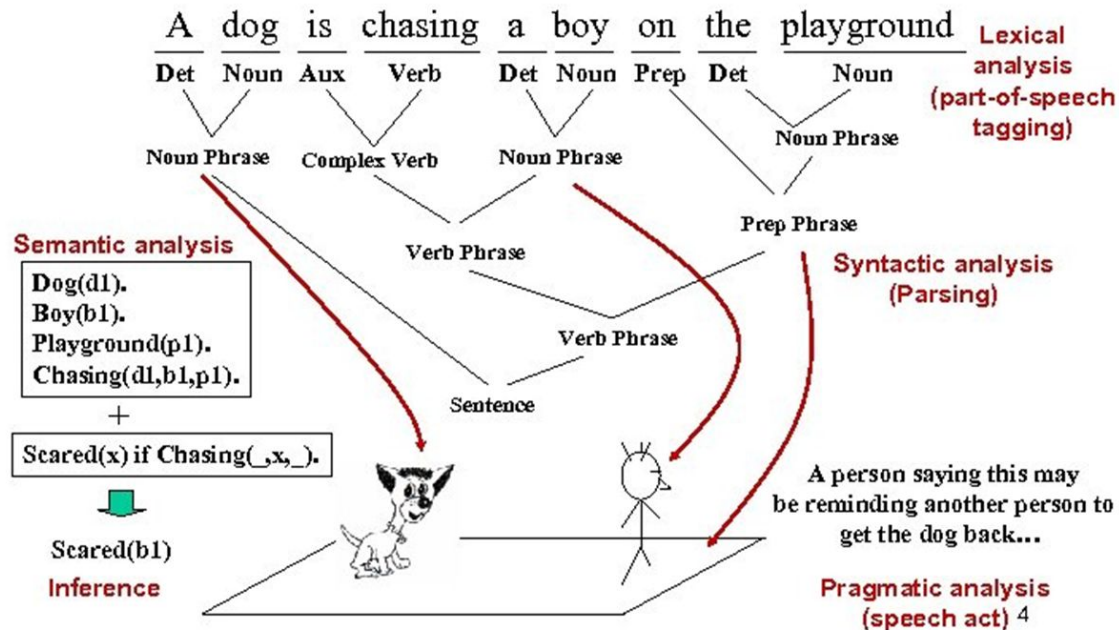
VES Institute of Technology, Mumbai

Agenda

- Pragmatic Analysis
- Discourse processing
- Reference Resolution,
- Reference Phenomena
- Syntactic & Semantic constraint on coherence
- Anaphora

An Example of NLP

An Example of NLP

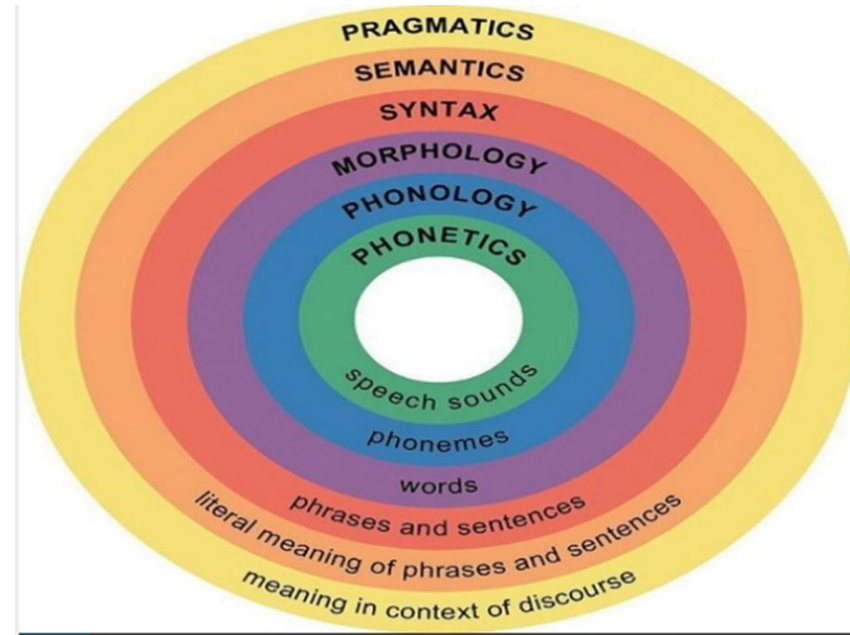


Pragmatics : Definition

- It is the **study of speaker meaning** : Concerned with study of meaning as communicated by speaker & interpreted by the listener.
- It is the **study of contextual meaning** : Involves interpretation of what people mean in a particular context and how the context influences what is said.
- It is the **study of how more gets communicated than is said** : Explores how great deal of what is unsaid is recognized as part of what is communicated.
- It is the **study of the expression of relative distance** : On the assumption of how close and distant the listener is speakers determine how much needs to be said.

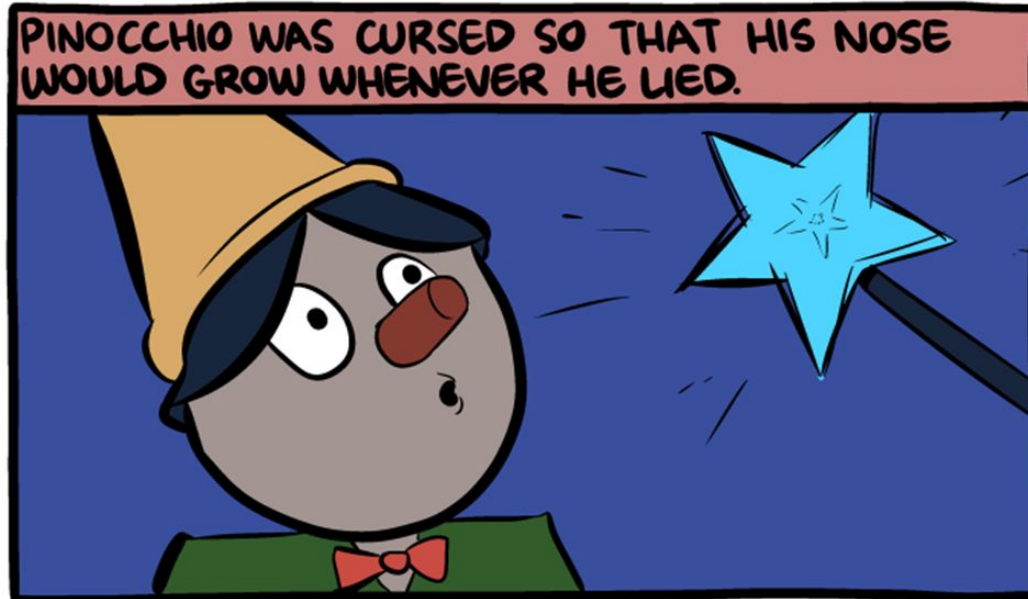
Pragmatics

	Grammar	Semantics	Pragmatics
Sound	Phonetics/ Phonology	---	---
Word	Morphology	Lexical Semantics	---
Sentence	Syntax	Compositional Semantics	Pragmatics
Text & Discourse	Text & Discourse Grammar	Discourse Semantics	Pragmatics
	Structure	Meaning	Use



Pragmatics : Definition

- **Study of the meaning of language in context :**
 - Literal Meaning (Semantics) → Meaning in Context



Pragmatics : Definition



Pragmatics : Phenomena

- **Three kinds of phenomena** : disparate analyses are all forms of '**associative meaning**':
 - (i) **extralinguistic-based** (e.g. the meaning associations of 'nurse' in English with 'female nurse' rather than 'male nurse'),
 - (ii) **linguistic-based** (e.g. **Fragrance vs. Odor (English)**): Both refer to a smell. However, *fragrance* is linguistically tethered to positive, aesthetic experiences (perfume, flowers), while *odor* has evolved a tendency to refer to biological or unpleasant smells (body odor, chemicals)); and
 - (iii) **Scalar implicature-based** (e.g. the tendency for 'some / most' to be interpreted as meaning 'not all').

Utterances, Sentences & Propositions

➤ **sentence**: an abstraction, the type of what was said

(1) Caesar invades Gaul

➤ **proposition**: a further abstraction, normally ignoring some non-literal meaning

(2) `invade(Caesar, Gaul)`

➤ **information structure**: what part of a proposition is emphasized

(3) Caesar invaded Gaul

(4) Gaul was invaded by Caesar

(5) It was Gaul that Caesar invaded

(6) It was Caesar who invaded Gaul

Pragmatics - Invisible meaning



Pragmatics - Invisible meaning



Coke addict.



In many ways
Pragmatics is the study of 'invisible' meaning. or **How we recognize what is meant** even when it isn't actually said.



Pragmatics - Definition

- **David Crystal's definition:** “Pragmatics studies the factors that govern our choice of language in social interaction and the effects of our choice on others.
- **George Keith's definition:** “Pragmatics is all about the meanings between the lexis and the grammar and the phonology. Meanings are implied and the rules being followed are unspoken, unwritten ones.”

Reference: [Link](#)

Pragmatics - Definition

Pragmatics tell **how language is used**; that is 'meaning in context'.

Example: if someone says "**the door is open**" then it is necessary to know which door "**the door**" refers to;

Need to know what the intention of the speaker :

- could be a pure **statement** of fact,

- could be an **explanation** of how the cat got in, or

- could be a **request** to the person addressed to close the door.



Five Aspects of Pragmatics

- Deixis
- Implicature
- Presupposition
- Speech Acts
- Conversational Structure

Deixis

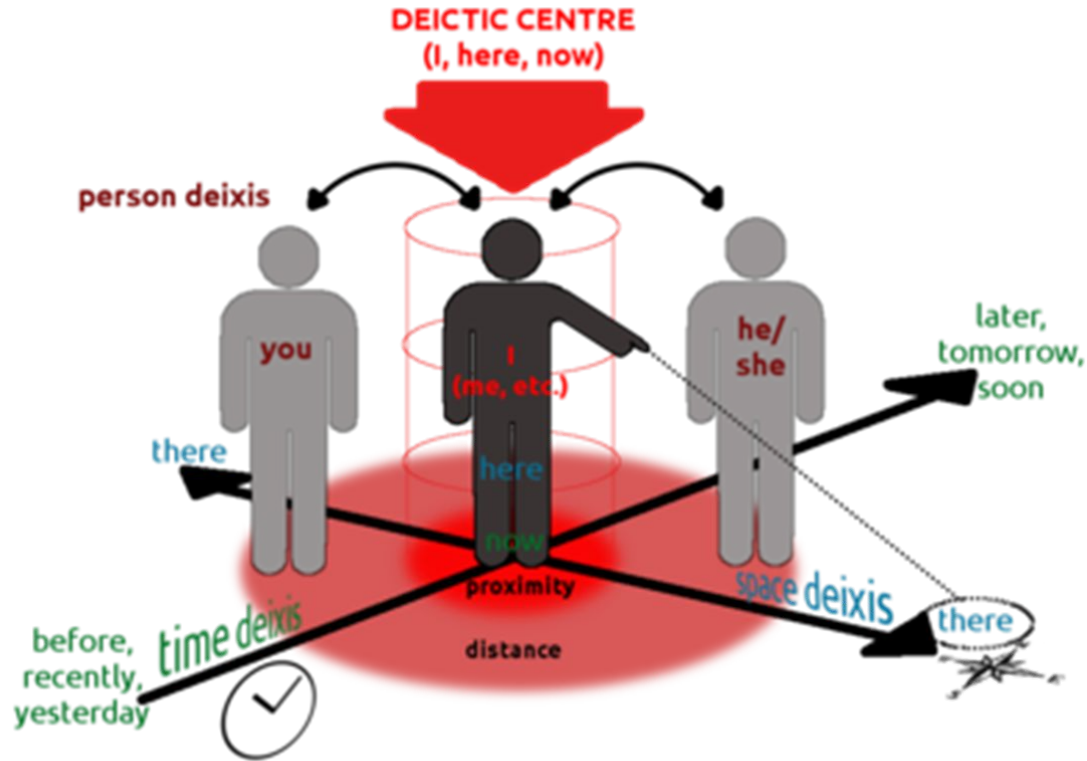
Deixis means words whose meaning depends on context.

- It concerns the ways in which language encode or grammaticalized features of the context of utterance or speech event
- Concerns ways in which the interpretation of utterances depends on the analysis of that context of utterance.
- It is a method of directly encoding context into language.

Example Scenario: Notice on Principal office door :

- *I'll be back in an hour*
- *Meet me after a break*
- *She called you to attend meeting*

Deixis



Implicature

- Implicature is a technical term , **refers to what is suggested in an utterance**, even though neither expressed nor strictly implied.
- **Example:** Ram is meeting a woman this evening
- **Implicature : The Woman** Ram is meeting this evening is not either his mother, sister nor wife.
- It **means more being communicated than is said**.

► Implicature

The aspect of meaning that a speaker conveys, implies, or suggests without directly expressing.

Can you pass the salt? may implicate “pass me the salt”

Presupposition

- Language often points to the **unstated commonly understood information**. We are able to use **inference** to find out what this **information** is.
- What **a speaker / writer assumes is true or known by a listener / reader** can be described as a presupposition.

Presupposition : Example

- **Ram's brother bought three elephants**

- **Presuppositions :**

Presupposition = information assumed to be true.

- Ram exists
- Ram has a brother
- Ram has only one brother
- He has a lot of money
- There are **both semantic and Pragmatic presuppositions** – involve shared knowledge between speaker and hearer.

Try it : Kim's spouse bakes the best pies

Speech Acts

- Austin (1962)
 - An utterance in dialogue is an ACTION – **Speech Acts**
 - **Speech acts**
 - Performative sentences uttered by an authority (they change the state of the world)
 - Any sentence in real speech contains / **Speech acts consists of three components:**
 - **Locutionary act** – utterance with particular meaning (the actual words which the speaker is saying);
 - **Illocutionary act** – asking, answering, promising, etc. (the intention of the speaker);
 - **Perlocutionary act** – effect upon feelings, thoughts, etc. (the effect of the utterance on the hearer).

Speech Acts : Classification

- Searle (1975) **All speech acts classified as**
- **Assertives** – suggesting, boasting, concluding, etc.
- **Directives** – asking, ordering, inviting, etc.
- **Commissives** – promising, planning, vowing, etc.
- **Expressives** – thanking, apologizing, deploring, etc.
- **Declarations** – performatives (state-changing)

Conversational Structure

- **Analysis of the sequential** (and anti-sequential) nature of conversations, interruptions, etc.
- **Conversational implicatures are context – dependant.** What is implied varies according to the context of an utterance.

Discourse Analysis : Introduction

- Discourse comes from Latin word 'discursus'
- **Discourse:** language consists of **collocated, related groups of sentences**. We **refer to such a group of sentences as a discourse**
- A continuous stretch of spoken language larger than a sentence, often constituting a **coherent** unit
- A stretch of language perceived to be meaningful unified, and **purposive; language in use** (viewed) as **social practice** determined by social structures.

Discourses of food
"Healthy Food"



Social Practice
Healthy lifestyle

Discourse Analysis : Structural Definition

- **Solving the problem:** adopt Lyons's distinction between **system-sentences** and **text – sentences**.
- System sentences are **well-formed abstract theoretical sentences** generated according to the existing grammar rules;
- Text-sentences are **context-dependent utterances** or parts of utterances which occur in everyday life.
- **The discourse analysis will be concerned with text-sentences.**

Discourse Analysis

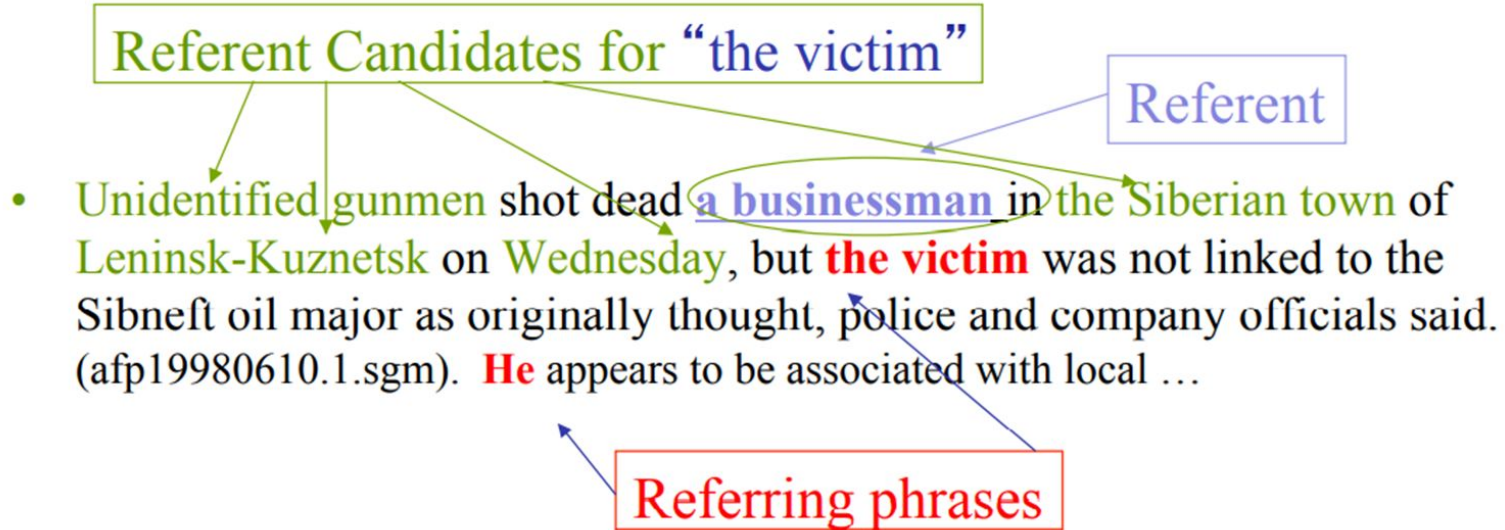
- There are **two basic forms** of discourse:
 - Monologue;
 - Dialogue;
- We will focus on techniques commonly applied to the interpretation of **monologues**.
- **Monologue** : The **communication flows in only one direction** in a monologue, i.e., from the speaker to the hearer.
- **Dialogue** : consist of a much **freer interchange**. In this case, each participant periodically takes turns being a speaker and hearer

Reference Resolution

- **Reference**: the process by which speakers use expressions to denote an entity. (E.g. Ram and he – denotes a person named Ram)
- **Referring expression**: natural language expression used to perform reference
- **Referent**: the entity that is referred to (Ram and he - are referring expressions, and Ram is their referent)
- **Coreference**: referring expressions that are used to refer to the same entity. (Ram and he are co-refer & **Ram is the antecedent of he**)
- **Anaphora**: reference to a previously introduced entity and the referring expression used is said to be **anaphoric**. (pronouns he and it are therefore anaphoric)

Terminology Examples

- The referent for a referring phrase is found by the resolution algorithm among the candidates, previous noun phrases.



Reference Resolution

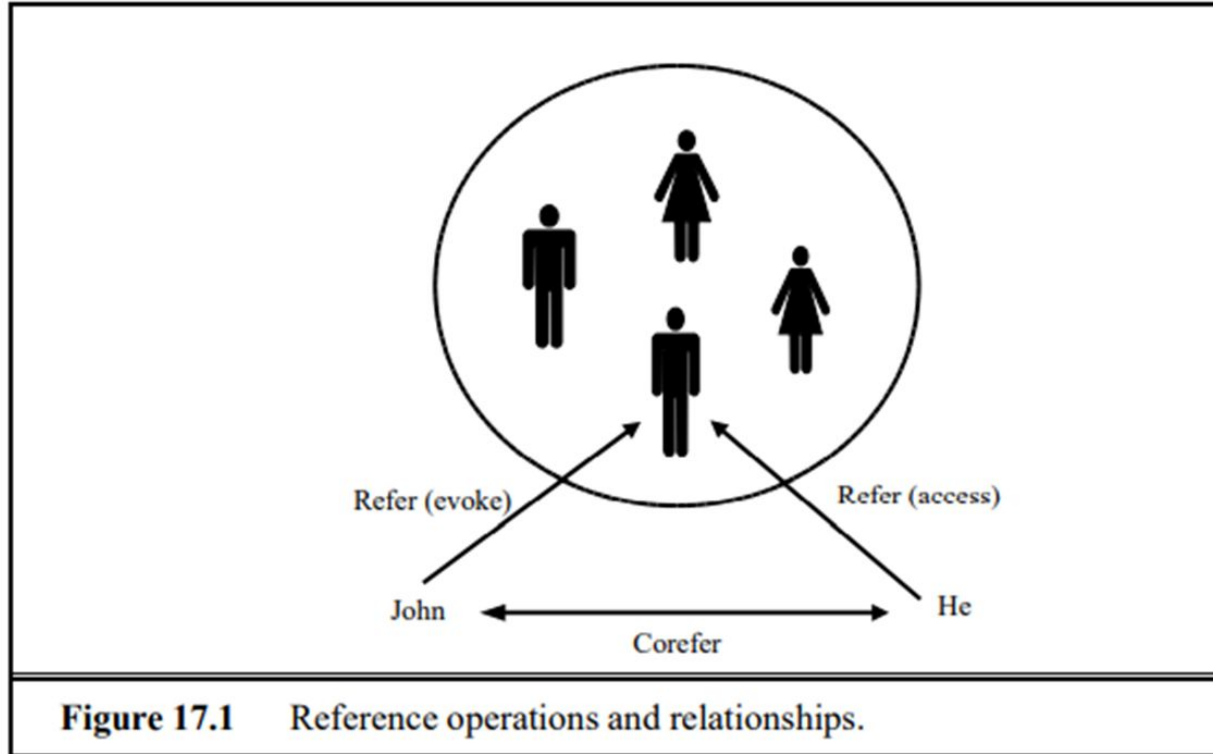
Discourse Model

- It contains **representations of the entities that have been referred to in the discourse and the relationships in which they participate.**

Two components required by a system to produce and interpret referring expressions.

- A method for **constructing a discourse model** that evolves dynamically.
- A method for **mapping between referring expressions and referents.**

Reference Operations & Relationships



Reference Operations & Relationships

According to John, Bob bought Sue an Integra, and Sue bought Fred a Legend.

- a. But that turned out to be a lie.
 - b. But that was false.
 - c. That struck me as a funny way to describe the situation.
 - d. That caused Sue to become rather poor.
 - e. That caused them both to become rather poor.
- The referent of that is a speech act in (a), a proposition in (b), a manner of description in (c), an event in (d), and a combination of several events in (e).
 - The field awaits the development of robust methods for interpreting these types of reference.

Reference Phenomena

Five common types of referring expression	
Type	Example
Indefinite noun phrase	I saw a Ford Escort today.
Definite noun phrase	I saw a Ford Escort today. The Escort was white.
Pronoun	I saw a Ford Escort today. It was white.
Demonstratives	I like this better than that .
One-anaphora	I saw 6 Ford Escort today. Now I want one .
Three types of referring expression that complicate the reference resolution	
Type	Example
Inferrables	I almost bought a Ford Escort, but a door had a dent.
Discontinuous Sets	John and Mary love their Escorts. They often drive them.
Generics	I saw 6 Ford Escorts today. They are the coolest cars.

Reference Phenomena : Pronoun

- **Pronouns** refer to entities that were introduced fairly recently
 - **Nominative** (he, she, it, they, etc.)
 - e.g. The German authorities said a Colombian₁ who had lived for a long time in the Ukraine flew in from Kiev. He₁ had 300 grams of plutonium 239 in his baggage.
 - **Oblique** (him, her, them, etc.)
 - e.g. Undercover investigators negotiated with three members of a criminal group₂ and arrested them₂ after receiving the first shipment.
 - **Possessive** (his, her, their, etc. + hers, theirs, etc.)
 - e.g. He₃ had 300 grams of plutonium 239 in his₃ baggage. The suspected smuggler₃* denied that the materials were his₃. (*chain)
 - **Reflexive** (himself, themselves, etc.)
 - e.g. There appears to be a growing problem of disaffected loners₄ who cut themselves₄ off from all groups .

Reference Phenomena

Names

- It is the simplest type of referring expression.
- It can be the name of a person, organization and location also.
- For example, Ram had gone around one day to bring him some food
- Ram is the name-referring expression.

Syntactic & Semantic Constraints on Coreference

- How to develop successful algorithms for reference resolution?
- There are **two necessary steps**.
- First is to **filter the set of possible referents** by certain hard-and-fast constraints.
- Second is to **set the preference for possible referents**.

Approach to Coreference Resolution

Identify Referent Candidates

- All noun phrases (both indef. and def.) are considered potential referent candidates.
- A referring phrase can also be a referent for a subsequent referring phrases,
 - Example: (omitted sentence with name of suspect)
He had 300 grams of plutonium 239 in **his** baggage. The suspected **smuggler** denied that the materials were **his**.
(chain of 4 referring phrases)
- All potential candidates are collected in a table collecting feature info on each candidate.
- Problems:
 - chunking
 - e.g. the Chase Manhattan Bank of New York
 - nesting of NPs

Constraints (English)

- A **categorization of pronouns with respect to number** is shown in Figure 17.2

Singular	Plural	Unspecified
she, her, he, him, his, it	we, us, they, them	you

Figure 17.2 Number agreement in the English pronominal system.

- A **categorization of pronouns with respect to person** is shown in Figure 17.3.

	First	Second	Third
Nominative	I, we	you	he, she, they
Accusative	me, us	you	him, her, them
Genitive	my, our	your	his, her, their

Figure 17.3 Person and case agreement in the English pronominal system.

masculine	feminine	nonpersonal
he, him, his	she, her	it

Figure 17.4 Gender agreement in the English pronominal system.

Constraints (English)

Number Agreement:

- To distinguish between singular and plural references.
- *John has a new car. They are red.

Gender Agreement:

- To distinguish male, female, and non-personal genders.
- John has a new car. It is attractive. [It = the new car]

Person and Case Agreement:

- To distinguish between three forms of person;
- *You and I have Escorts. They love them.
- To distinguish between subject position, object position, and genitive position.

Constraints (English)

Syntactic Constraints:

- Syntactic relationships between a referring expression and a possible antecedent noun phrase
 - John bought himself a new car. [himself=John]
 - John bought him a new car. [him≠John]
- English pronouns such as **himself**, **herself**, and **themselves** are called **Reflexives**

Selectional Restrictions:

- A verb places restrictions on its arguments.
 - John parked his Acura in the garage. He had driven it around for hours. [it=Acura, it≠garage];
 - I picked up the book and sat in a chair. It broke

Preferences in Pronoun Interpretation

Recency:

- Entities introduced recently are more salient than those introduced before.
 - John has a Legend. Bill has an Escort. Mary likes to drive it.

Grammatical Role:

- Entities mentioned in subject position are more salient than those in object position.
 - Bill went to the Acura dealership with John. He bought an Escort. [he=Bill]
 - John went to the Acura dealership with Bill. He bought an Integra. [he = John]
 - John and Bill went to the Acura dealership. He bought an Integra. [he = ??].

Preferences in Pronoun Interpretation

Repeated Mention:

- Entities that have been focused on in the prior discourse are more salient.
 - John needed a car to get to his new job.
 - He decided that he wanted something sporty.
 - Bill went to the Acura dealership with him.
 - He bought an Integra. [he=John]

Preferences in Pronoun Interpretation

Parallelism (more generally – discourse structure):

- There are also strong preferences that appear to be induced by parallelism effects.
 - Mary went with Sue to the cinema. Sally went with her to the mall. [her = Sue]
 - Mary went with Sue to the Acura dealership. Sally told her not to buy anything. [her = Mary]
 - Jim surprised Paul and then Julie shocked him. (*him* = Paul)

Preferences in Pronoun Interpretation

Verb Semantics:

- Certain verbs appear to place a semantically-oriented emphasis on one of their argument positions.
 - John telephoned Bill. He had lost the book in the mall. [He = John]
 - John criticized Bill. He had lost the book in the mall. [He = Bill]
 - David praised Hans because he ... [he = Hans]
 - David apologized to Hans because he... [he = David]

Anaphora Resolution Tools

- ▶ OPEN NLP
 - <http://opennlp.sourceforge.net>
- ▶ CherryPicker: A Coreference Resolution Tool
 - <http://www.hlt.utdallas.edu/~altaf/cherrypicker.html>
- ▶ Reconcile – Coreference Resolution Engine
 - <http://www.cs.utah.edu/nlp/reconcile/>

Anaphora Resolution Tools

▶ GUITAR

- <http://cswww.essex.ac.uk/Research/nle/GuiTAR/gtarNew.html>
- Uses Charniak's parser
- Input Minimally Associated XML format (MAS-XML)
- Output in XML
- Java

▶ BART

- <http://www.bart-coref.org/>
- John Hopkins University Tool for AR
- Uses Charniak's parser
- Comes up with Machine learning model
- Can take Input as XML. Output XML

▶ MARS

- <http://clg.wlv.ac.uk/demos/MARS/index.php>
- Built on Mitkov principle
- Uses certain linguistic rules
- If your data has noises, don't use it.

▶ Java-RAP (pronouns)

- <http://aye.comp.nus.edu.sg/~qiu/NLPTools/JavaRAP.html>
- Working on Lipin and Leass rules (1994)
- Uses Charniak's parser